21-242 Recitation 12

Midterm 1 Review Solutions

Solutions at the level of each question,
with later-course alternatives where they help

Thursday, October 1, 2026

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These notes follow the recitation handout in order. Each item has a **course-level** writeup using only ideas that are natural at that point in the course. When a later theorem (coordinates, matrices, rank-nullity, dimension counting) makes the same claim shorter or more conceptual, a **later-course** writeup is included.

Throughout, V and W are vector spaces over a field $\mathbb{F}$, unless a question is specifically about matrices $A \in \mathbb{F}^{m\times n}$. Lists may be empty; the empty sum is $\vec{0}$.

1 Fields and Vector Spaces

1.1 Q1. Unique additive identity

Prompt

State and prove that $\vec{0} \in V$ is the unique additive identity.

Course-level solution

Statement. There is a unique vector $\vec{0} \in V$ such that $\vec{v} + \vec{0} = \vec{v}$ for every $\vec{v} \in V$.
Existence is an axiom of a vector space.
Uniqueness. Suppose $\vec{e} \in V$ also satisfies $\vec{v} + \vec{e} = \vec{v}$ for every $\vec{v} \in V$. Taking $\vec{v} = \vec{0}$ and using that $\vec{0}$ is an additive identity gives

$$\vec{e} = \vec{0} + \vec{e} = \vec{0}.$$

Thus the additive identity is unique.

Later-course solution (using ideas from the rest of the handout)

The same one-line uniqueness argument works in any abelian group, so it is not special to vector spaces. After you know that $V \cong \mathbb{F}^n$ when $\dim V = n < \infty$, uniqueness of $\vec{0}$ in V is equivalent to uniqueness of the zero tuple in $\mathbb{F}^n$, but that is a strictly heavier route and hides that the claim is purely algebraic.

1.2 Q2. $0\vec{v} = \vec{0}$

Prompt

Prove that $0\vec{v} = \vec{0}$.

Course-level solution

Let $\vec{v} \in V$. Then

$$0\vec{v} = (0 + 0)\vec{v} = 0\vec{v} + 0\vec{v}$$

by distributivity. Add the additive inverse $-(0\vec{v})$ to both sides:

$$\vec{0} = 0\vec{v}.$$

(Equivalently: $0\vec{v} + 0\vec{v} = 0\vec{v}$ and Q4 below, or cancellation.)

Later-course solution (using ideas from the rest of the handout)

In coordinates relative to any basis $\mathcal{B}$, we have $[0\vec{v}]_{\mathcal{B}} = 0[\vec{v}]_{\mathcal{B}} = \vec{0}$ in $\mathbb{F}^n$, hence $0\vec{v} = \vec{0}$. This uses the coordinate isomorphism, which is later in the handout.

1.3 Q3. $-\vec{v} = (-1)\vec{v}$

Prompt

Prove that $-\vec{v} = (-1)\vec{v}$.

Course-level solution

It is enough to check that $(-1)\vec{v}$ is an additive inverse of $\vec{v}$, because additive inverses are unique (if $\vec{v}+\vec{w} = \vec{0} = \vec{v}+\vec{u}$ then $\vec{w} = \vec{0}+\vec{w} = (-\vec{v}+\vec{v})+\vec{w} = -\vec{v}+(\vec{v}+\vec{w}) = -\vec{v}+\vec{0} = -\vec{v}$, and likewise $\vec{u} = -\vec{v}$).
Now

$$\vec{v} + (-1)\vec{v} = 1\cdot\vec{v} + (-1)\vec{v} = \big(1+(-1)\big)\vec{v} = 0\vec{v} = \vec{0},$$

using Q2. Therefore $-\vec{v} = (-1)\vec{v}$.

Later-course solution (using ideas from the rest of the handout)

The coordinate proof is again $[\vec{v}] + [(-1)\vec{v}] = [\vec{v}] + (-1)[\vec{v}] = \vec{0}$ in $\mathbb{F}^n$. The axiom-level proof is the one you want on a midterm at this stage of the course.

1.4 Q4. Cancellation / $\vec{v}+\vec{w} = \vec{v}$ implies $\vec{w} = \vec{0}$

Prompt

Show that if $\vec{v}+\vec{w} = \vec{v}$, then $\vec{w} = \vec{0}$.

Course-level solution

Add $-\vec{v}$ to both sides:

$$-\vec{v}+(\vec{v}+\vec{w}) = -\vec{v}+\vec{v} \implies (-\vec{v}+\vec{v})+\vec{w} = \vec{0} \implies \vec{0}+\vec{w} = \vec{0} \implies \vec{w} = \vec{0}.$$

Later-course solution (using ideas from the rest of the handout)

This is the cancellation law in the abelian group $(V,+)$. Equivalently, the translation map $T_{\vec{v}}(\vec{w}) = \vec{v}+\vec{w}$ is injective (in fact invertible, with inverse $T_{-\vec{v}}$).

1.5 Q5. $\vec{v} = -\vec{v}$ need not imply $\vec{v} = \vec{0}$

Prompt

Show that $\vec{v} = -\vec{v}$ does not necessarily imply $\vec{v} = \vec{0}$.

Course-level solution

The equation $\vec{v} = -\vec{v}$ is equivalent to $2\vec{v} = \vec{0}$. If $1 + 1 \neq 0$ in $\mathbb{F}$ (characteristic not 2), we may multiply by the scalar 2^{-1} and conclude $\vec{v} = \vec{0}$. So a counterexample must live in characteristic 2.
Take $\mathbb{F} = \mathbb{F}_2 = \{0, 1\}$ with $1 + 1 = 0$, and take $V = \mathbb{F}_2$ as a vector space over itself. Let $\vec{v} = 1$. Then

$$-\vec{v} = (-1)\vec{v} = 1 \cdot 1 = 1 = \vec{v},$$

but $\vec{v} \neq \vec{0}$.
(The same phenomenon occurs in any vector space over a field of characteristic 2: every vector satisfies $\vec{v} = -\vec{v}$.)

Later-course solution (using ideas from the rest of the handout)

Once you have bases: if $\operatorname{char} \mathbb{F} = 2$ and $\dim V \geq 1$, any nonzero basis vector is a counterexample. If $\operatorname{char} \mathbb{F} \neq 2$, then $\vec{v} = -\vec{v}$ really does imply $\vec{v} = \vec{0}$, so the hypothesis "not necessarily" is a characteristic-2 statement, not a dimension statement.

2 Span, Linear Independence, Bases

We take the following as the working definitions at this point of the course (Axler / 21-242 style). A *linear combination* of a list $\mathcal{L} = (\vec{v}_1, \ldots, \vec{v}_n)$ is a vector $\sum_{i=1}^{n} a_i \vec{v}_i$ with $a_i \in \mathbb{F}$.

2.1 Q1. Definition of $\operatorname{Span}(\mathcal{L})$

Prompt

State the definition of $\operatorname{Span}(\mathcal{L})$.

Course-level solution

Definition. If $\mathcal{L} = (\vec{v}_1, \ldots, \vec{v}_n)$ is a list of vectors in V, then

$$\operatorname{Span}(\mathcal{L}) := \left\{ \sum_{i=1}^{n} a_i \vec{v}_i : a_1, \ldots, a_n \in \mathbb{F} \right\}.$$

If $\mathcal{L}$ is empty, $\operatorname{Span}(\mathcal{L}) = \{\vec{0}\}$.

Equivalently: $\operatorname{Span}(\mathcal{L})$ is the set of all linear combinations of vectors appearing in $\mathcal{L}$. It is the smallest subspace of V containing every vector of $\mathcal{L}$.

2.2 Q2. Definition of linear independence

Prompt

State the definition of linear independence.

Course-level solution

Definition (Axler / sequential form)**.** A list $(\vec{v}_1, \ldots, \vec{v}_n)$ in V is *linearly independent* if no vector in the list is a linear combination of the preceding ones: that is, $\vec{v}_1 \neq \vec{0}$ and $\vec{v}_j \notin \text{Span}(\vec{v}_1, \ldots, \vec{v}_{j-1})$ for each $j = 2, \ldots, n$. The empty list is linearly independent.

A list is *linearly dependent* if it is not linearly independent.
(An equivalent and often more convenient form is Q3 below.)

2.3 Q3. Independence iff only the trivial combination is $\vec{0}$

Prompt

Prove that $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n \subseteq V$ is linearly independent if and only if $\sum_{i=1}^{n} a_i \vec{v}_i = \vec{0}$ implies $a_i = 0$ for all i.

Course-level solution

Write $\mathcal{L} = (\vec{v}_1, \ldots, \vec{v}_n)$. We use the sequential definition from Q2.
($\Rightarrow$). Suppose $\mathcal{L}$ is linearly independent and $\sum_{i=1}^{n} a_i \vec{v}_i = \vec{0}$. If some coefficient is nonzero, let j be the *largest* index with $a_j \neq 0$. Then

$$\vec{v}_j = \sum_{i=1}^{j-1} \Big(-\frac{a_i}{a_j}\Big) \vec{v}_i \in \text{Span}(\vec{v}_1, \ldots, \vec{v}_{j-1})$$

(if $j = 1$ this says $\vec{v}_1 = \vec{0}$). This contradicts sequential independence. Hence every $a_i = 0$.
($\Leftarrow$). Suppose the only linear combination equal to $\vec{0}$ is the trivial one. If the list failed sequential independence, some $\vec{v}_j$ would lie in $\text{Span}(\vec{v}_1, \ldots, \vec{v}_{j-1})$, say $\vec{v}_j = \sum_{i<j} c_i \vec{v}_i$. Then

$$\sum_{i<j} c_i \vec{v}_i + (-1)\vec{v}_j + \sum_{i>j} 0 \cdot \vec{v}_i = \vec{0}$$

is a nontrivial combination (the coefficient of $\vec{v}_j$ is -1), a contradiction. Thus $\mathcal{L}$ is linearly independent.

Later-course solution (using ideas from the rest of the handout)

After bases and dimension: a list of length n in an n-dimensional space is independent iff it is spanning iff it is a basis, and the trivial-combination criterion is the statement that the map $\mathbb{F}^n \to V$, $e_i \mapsto \vec{v}_i$, is injective. That is a cleaner packaging, but it uses the linear-map language from the next section.

2.4 Q4. Can $\vec{0}$ appear in a linearly independent list?

Prompt

Can $\vec{0}$ appear in a linearly independent list?

Course-level solution

No. If $\vec{v}_j = \vec{0}$, then $1 \cdot \vec{v}_j = \vec{0}$ is a nontrivial linear combination of the list (or, in sequential language, $\vec{0}$ always lies in the span of the preceding vectors, including the empty span when $j = 1$).
So any list containing $\vec{0}$ is linearly dependent.

2.5 Q5. Every list contains a basis of its span

Prompt

Prove that for any list $\mathcal{L} \subseteq V$, there exists a sublist $\mathcal{L}' \subseteq \mathcal{L}$ such that $\mathcal{L}'$ is a basis for $\text{Span}(\mathcal{L})$.

Course-level solution

Let $\mathcal{L} = (\vec{v}_1, \ldots, \vec{v}_n)$. Build $\mathcal{L}'$ by reading $\mathcal{L}$ from left to right and keeping a vector if and only if it is not in the span of those already kept. (Equivalently: drop each vector that is a linear combination of the preceding vectors of $\mathcal{L}$.)

- $\mathcal{L}'$ is linearly independent by construction (sequential definition).
- $\text{Span}(\mathcal{L}') = \text{Span}(\mathcal{L})$: every dropped vector already lay in the span of earlier kept vectors, so by induction each $\vec{v}_j$ lies in $\text{Span}(\mathcal{L}')$. The reverse inclusion is obvious.

A linearly independent spanning list of $\text{Span}(\mathcal{L})$ is a basis of $\text{Span}(\mathcal{L})$.

Later-course solution (using ideas from the rest of the handout)

Once you have the extension lemma and dim: take a maximal linearly independent sublist $\mathcal{L}' \subseteq \mathcal{L}$. Maximality implies it spans $\text{Span}(\mathcal{L})$, else you could add another vector of $\mathcal{L}$. Finite generation of the span is what makes maximality work without Zorn.

3 Linear Transformations

3.1 Q1. Definition

Prompt

State the definition of a linear transformation.

Course-level solution

Definition. A map $T : V \to W$ is a *linear transformation* if

$$T(a\vec{v} + b\vec{w}) = a\,T(\vec{v}) + b\,T(\vec{w}) \qquad \text{for all } \vec{v}, \vec{w} \in V \text{ and all } a, b \in \mathbb{F}.$$

Equivalently: $T(\vec{v} + \vec{w}) = T(\vec{v}) + T(\vec{w})$ and $T(a\vec{v}) = a\,T(\vec{v})$.

3.2 Q2. $T(\vec{0}_V) = \vec{0}_W$

Prompt

Prove that $T(\vec{0}_V) = \vec{0}_W$.

Course-level solution

$$T(\vec{0}_V) = T(\vec{0}_V + \vec{0}_V) = T(\vec{0}_V) + T(\vec{0}_V).$$

Cancellation (Fields Q4) yields $T(\vec{0}_V) = \vec{0}_W$.
Alternatively, $T(\vec{0}_V) = T(0 \cdot \vec{0}_V) = 0 \cdot T(\vec{0}_V) = \vec{0}_W$.

3.3 Q3. Unique linear map sending a basis to prescribed vectors

Prompt

If $\vec{v}_1, \ldots, \vec{v}_n$ are a basis for V and $\vec{w}_1, \ldots, \vec{w}_n \in W$, there exists a unique linear transformation $T : V \to W$ with $T(\vec{v}_i) = \vec{w}_i$ for all $i \in [n]$.

Course-level solution

Existence. Each $\vec{v} \in V$ may be written uniquely as $\vec{v} = \sum_{i=1}^{n} a_i \vec{v}_i$. Define

$$T(\vec{v}) := \sum_{i=1}^{n} a_i \vec{w}_i.$$

If $\vec{v} = \sum a_i \vec{v}_i$ and $\vec{u} = \sum b_i \vec{v}_i$, then $c\vec{v} + d\vec{u} = \sum (ca_i + db_i)\vec{v}_i$, so

$$T(c\vec{v} + d\vec{u}) = \sum (ca_i + db_i)\vec{w}_i = c\,T(\vec{v}) + d\,T(\vec{u}).$$

By construction $T(\vec{v}_i) = \vec{w}_i$.
Uniqueness. If S is linear and $S(\vec{v}_i) = \vec{w}_i$, then for $\vec{v} = \sum a_i \vec{v}_i$,

$$S(\vec{v}) = \sum a_i S(\vec{v}_i) = \sum a_i \vec{w}_i = T(\vec{v}).$$

Later-course solution (using ideas from the rest of the handout)

In matrix language (next section): the unique T is the map whose matrix $[T]_{\mathcal{CB}}$ has jth column $[\vec{w}_j]_{\mathcal{C}}$, for any basis $\mathcal{C}$ of W. Uniqueness is the statement that a linear map is determined by

its matrix.

3.4 Q4. Coordinates $[\vec{v}]_{\mathcal{B}}$

Prompt

Define $[\vec{v}]_{\mathcal{B}}$ (where $\mathcal{B}$ is a basis for V) and prove that $\vec{v} \mapsto [\vec{v}]_{\mathcal{B}}$ is an invertible linear transformation.

Course-level solution

Let $\mathcal{B} = (\vec{b}_1, \ldots, \vec{b}_n)$ be a basis of V. Every $\vec{v} \in V$ may be written uniquely as $\vec{v} = \sum_{i=1}^{n} a_i \vec{b}_i$. Define

$$[\vec{v}]_{\mathcal{B}} := \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{F}^n.$$

Let $\Phi_{\mathcal{B}} : V \to \mathbb{F}^n$ by $\Phi_{\mathcal{B}}(\vec{v}) = [\vec{v}]_{\mathcal{B}}$.
Linearity. If $\vec{v} = \sum a_i \vec{b}_i$ and $\vec{w} = \sum c_i \vec{b}_i$, then $r\vec{v} + s\vec{w} = \sum (ra_i + sc_i)\vec{b}_i$, so $[\, r\vec{v} + s\vec{w} \,]_{\mathcal{B}} = r[\vec{v}]_{\mathcal{B}} + s[\vec{w}]_{\mathcal{B}}$.
Invertibility. $\Phi_{\mathcal{B}}$ is bijective because each coordinate tuple corresponds to exactly one linear combination of $\mathcal{B}$. Explicitly, the inverse is

$$\Phi_{\mathcal{B}}^{-1} : \mathbb{F}^n \to V, \qquad \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \mapsto \sum_{i=1}^{n} a_i \vec{b}_i,$$

which is linear (it is the unique map sending the standard basis vector e_i to $\vec{b}_i$, by Q3). Thus $\Phi_{\mathcal{B}}$ is an invertible linear transformation, i.e. an isomorphism $V \cong \mathbb{F}^n$.

3.5 Q5. Surjectivity iff the image of a basis spans

Prompt

Let $\mathcal{B} \subseteq V$ be a basis. Prove that $T : V \to W$ is surjective iff $T(\vec{b}_1), \ldots, T(\vec{b}_n)$ spans W.

Course-level solution

Let $\mathcal{B} = (\vec{b}_1, \ldots, \vec{b}_n)$. For any $\vec{v} = \sum a_i \vec{b}_i$,

$$T(\vec{v}) = \sum a_i T(\vec{b}_i),$$

so

$$\operatorname{im}(T) = \operatorname{Span}\big(T(\vec{b}_1), \ldots, T(\vec{b}_n)\big).$$

Therefore T is surjective iff $\operatorname{im}(T) = W$ iff $T(\vec{b}_1), \ldots, T(\vec{b}_n)$ spans W.

3.6 Q6. Injectivity iff the image of a basis is independent

Prompt

Prove that $T : V \to W$ is injective iff $T(\vec{b}_1), \ldots, T(\vec{b}_n)$ is linearly independent.

Course-level solution

($\Rightarrow$). Suppose T is injective and $\sum a_i T(\vec{b}_i) = \vec{0}$. Then $T\big(\sum a_i \vec{b}_i\big) = \vec{0}$, so $\sum a_i \vec{b}_i \in \ker T = \{\vec{0}\}$ (injective linear maps have trivial kernel: if $T(\vec{v}) = T(\vec{w})$ then $T(\vec{v} - \vec{w}) = \vec{0}$, so $\vec{v} - \vec{w} = \vec{0}$). Independence of $\mathcal{B}$ forces every $a_i = 0$. Thus $\big(T(\vec{b}_i)\big)$ is independent.
($\Leftarrow$). Suppose $\big(T(\vec{b}_i)\big)$ is independent and $T(\vec{v}) = \vec{0}$. Write $\vec{v} = \sum a_i \vec{b}_i$. Then $\sum a_i T(\vec{b}_i) = \vec{0}$, so every $a_i = 0$, hence $\vec{v} = \vec{0}$. Thus $\ker T = \{\vec{0}\}$, so T is injective.

Later-course solution (using ideas from the rest of the handout)

After rank-nullity: T is injective iff $\dim \ker T = 0$ iff $\dim \operatorname{im}(T) = \dim V = n$ iff the n vectors $T(\vec{b}_i)$ are independent in W (an independent list of length $\dim \operatorname{im}(T)$ is a basis of the image).

3.7 Q7. Bijectivity iff the image of a basis is a basis

Prompt

Prove that $T : V \to W$ is bijective iff $T(\vec{b}_1), \ldots, T(\vec{b}_n)$ is a basis for W.

Course-level solution

T is bijective iff T is injective and surjective iff $\big(T(\vec{b}_i)\big)$ is linearly independent (Q6) and spanning (Q5) iff $\big(T(\vec{b}_i)\big)$ is a basis of W.

Later-course solution (using ideas from the rest of the handout)

If you already know $\dim W = n = \dim V$, any two of independent / spanning / basis for the list $\big(T(\vec{b}_i)\big)$ are equivalent, matching the later fact that injectivity, surjectivity, and bijectivity coincide when $\dim V = \dim W < \infty$.

4 Matrices

Let $T : V \to W$ be linear, let $\mathcal{B} = (\vec{b}_1, \ldots, \vec{b}_n)$ be a basis of V and $\mathcal{C} = (\vec{c}_1, \ldots, \vec{c}_m)$ a basis of W. Write $\mathrm{Mat}_{m\times n}(\mathbb{F})$ for $m \times n$ matrices over $\mathbb{F}$, and $\mathcal{L}(V, W)$ for linear maps $V \to W$.

4.1 Q1. Definition of $[T]_{\mathcal{CB}}$ and the map $T \mapsto [T]_{\mathcal{CB}}$

Prompt

Define $[T]_{\mathcal{CB}}$ and show that $T \mapsto [T]_{\mathcal{CB}}$ is an invertible linear transformation.

Course-level solution

Definition. The *matrix of T with respect to $\mathcal{B}$ and $\mathcal{C}$* is the $m \times n$ matrix $[T]_{\mathcal{CB}}$ whose jth column is $[T(\vec{b}_j)]_{\mathcal{C}}$.

Let $\Psi : \mathcal{L}(V, W) \to \mathrm{Mat}_{m\times n}(\mathbb{F})$ by $\Psi(T) = [T]_{\mathcal{CB}}$.
Linearity. The jth column of $[S+T]_{\mathcal{CB}}$ is $[(S+T)(\vec{b}_j)]_{\mathcal{C}} = [S(\vec{b}_j)]_{\mathcal{C}} + [T(\vec{b}_j)]_{\mathcal{C}}$, and likewise $[aT]_{\mathcal{CB}} = a[T]_{\mathcal{CB}}$. So Ψ is linear.
Inverse. Given $A \in \mathrm{Mat}_{m\times n}(\mathbb{F})$ with columns $\vec{a}_1, \ldots, \vec{a}_n \in \mathbb{F}^m$, let $\vec{w}_j$ be the unique vector in W with $[\vec{w}_j]_{\mathcal{C}} = \vec{a}_j$, and let T be the unique linear map with $T(\vec{b}_j) = \vec{w}_j$ (Linear Transformations Q3). Then $[T]_{\mathcal{CB}} = A$. Uniqueness of T for a given matrix is the same construction in reverse. Thus Ψ is bijective, hence an invertible linear map: $\mathcal{L}(V, W) \cong \mathrm{Mat}_{m\times n}(\mathbb{F})$.

Later-course solution (using ideas from the rest of the handout)

Once dimensions are in hand, $\dim \mathcal{L}(V, W) = (\dim V)(\dim W) = nm = \dim \mathrm{Mat}_{m\times n}(\mathbb{F})$, so it is enough to check that Ψ is injective: $[T]_{\mathcal{CB}} = 0$ means $T(\vec{b}_j) = \vec{0}$ for all j, so $T = 0$.

4.2 Q2. Every matrix is some $[T]_{\mathcal{CB}}$

Prompt

Show that every matrix A can be written as $A = [T]_{\mathcal{CB}}$ for some linear transformation T and bases $\mathcal{C}$ and $\mathcal{B}$.

Course-level solution

Let $A \in \mathbb{F}^{m\times n}$. Take $V = \mathbb{F}^n$, $W = \mathbb{F}^m$, let $\mathcal{B}$ and $\mathcal{C}$ be the standard bases, and let $T = T_A$ (the map $\vec{x} \mapsto A\vec{x}$; defined in Q5). The jth column of A is $Ae_j = T_A(e_j)$, whose standard coordinates are the jth column of A. Thus $[T_A]_{\mathcal{CB}} = A$.

Later-course solution (using ideas from the rest of the handout)

You may also keep V and W arbitrary of dimensions n and m, choose any bases $\mathcal{B}, \mathcal{C}$, and take the unique T with $[T]_{\mathcal{CB}} = A$ (Q1). So every matrix arises in every pair of spaces of the right dimensions.

4.3 Q3. $[T]_{\mathcal{CB}}[\vec{v}]_{\mathcal{B}} = [T(\vec{v})]_{\mathcal{C}}$

Prompt

Prove that $[T]_{\mathcal{CB}}[\vec{v}]_{\mathcal{B}} = [T(\vec{v})]_{\mathcal{C}}$.

Course-level solution

Write $\vec{v} = \sum_{j=1}^{n} x_j \vec{b}_j$, so $[\vec{v}]_{\mathcal{B}} = \vec{x} = (x_1, \ldots, x_n)^T$. Then $T(\vec{v}) = \sum_{j=1}^{n} x_j T(\vec{b}_j)$, hence

$$[T(\vec{v})]_{\mathcal{C}} = \sum_{j=1}^{n} x_j [T(\vec{b}_j)]_{\mathcal{C}} = [T]_{\mathcal{CB}}\, \vec{x} = [T]_{\mathcal{CB}}\, [\vec{v}]_{\mathcal{B}},$$

because multiplying a matrix by $\vec{x}$ is the corresponding linear combination of its columns.

4.4 Q4. Composition corresponds to matrix multiplication

Prompt

Prove that $[S \circ T]_{\mathcal{CA}} = [S]_{\mathcal{CB}}[T]_{\mathcal{BA}}$.

Course-level solution

Let $T : U \to V$ and $S : V \to W$, with bases $\mathcal{A}$ of U, $\mathcal{B}$ of V, $\mathcal{C}$ of W. For any $\vec{u} \in U$, Q3 three times gives

$$\begin{aligned}[S \circ T]_{\mathcal{CA}}\, [\vec{u}]_{\mathcal{A}} &= [(S \circ T)(\vec{u})]_{\mathcal{C}} = [S(T(\vec{u}))]_{\mathcal{C}} = [S]_{\mathcal{CB}}[T(\vec{u})]_{\mathcal{B}} \\ &= [S]_{\mathcal{CB}}[T]_{\mathcal{BA}}\, [\vec{u}]_{\mathcal{A}}.\end{aligned}$$

The coordinate map $U \to \mathbb{F}^{\dim U}$ is surjective, so two matrices that act the same on every coordinate vector are equal: $[S \circ T]_{\mathcal{CA}} = [S]_{\mathcal{CB}}[T]_{\mathcal{BA}}$.
(Directly on columns: the jth column of the left side is $[(S \circ T)(a_j)]_{\mathcal{C}} = [S]_{\mathcal{CB}}[T(a_j)]_{\mathcal{B}}$, which is the jth column of the product.)

4.5 Q5. Define T_A

Prompt

Define T_A.

Course-level solution

Definition. For $A \in \mathbb{F}^{m \times n}$, the associated linear map is

$$T_A : \mathbb{F}^n \to \mathbb{F}^m, \qquad T_A(\vec{x}) := A\vec{x}.$$

Its standard matrix is A itself. Kernel and image of T_A are the nullspace and column space of A.

4.6 Q6. When is $[T_A]_{\mathcal{CB}} = A$ for every A?

Prompt

For which $\mathcal{C}, \mathcal{B}$ can we say for all A that $[T_A]_{\mathcal{CB}} = A$?

Course-level solution

Here $T_A : \mathbb{F}^n \to \mathbb{F}^m$, so $\mathcal{B}$ is a basis of $\mathbb{F}^n$ and $\mathcal{C}$ is a basis of $\mathbb{F}^m$. Let P be the matrix with columns the $\mathcal{B}$-vectors (so P is the change-of-basis matrix from $\mathcal{B}$-coordinates to standard coordinates: $\vec{x} = P[\vec{x}]_{\mathcal{B}}$), and let Q be the matrix with columns the $\mathcal{C}$-vectors. Then

$$T_A(\vec{x}) = A\vec{x} \implies Q[T_A(\vec{x})]_{\mathcal{C}} = AP[\vec{x}]_{\mathcal{B}},$$

hence $[T_A]_{\mathcal{CB}} = Q^{-1}AP$.
We need $Q^{-1}AP = A$ for *every* $A \in \mathbb{F}^{m\times n}$. Taking $A = I$ (when $m = n$) already forces $Q^{-1}P = I$, so $P = Q$; then $Q^{-1}AQ = A$ for all A forces Q to commute with every matrix, hence Q is scalar, but $Q^{-1}AQ = A$ for all A in the rectangular case as well:
Take $A = E_{ij}$ (matrix units). Then $Q^{-1}E_{ij}P = E_{ij}$ for all i, j implies $P = I_n$ and $Q = I_m$.
Thus $[T_A]_{\mathcal{CB}} = A$ for all A if and only if $\mathcal{B}$ is the standard basis of $\mathbb{F}^n$ and $\mathcal{C}$ is the standard basis of $\mathbb{F}^m$.

Later-course solution (using ideas from the rest of the handout)

In one line: $[T_A]_{\mathcal{CB}} = Q^{-1}AP$, and the only pair of invertible matrices satisfying $Q^{-1}AP = A$ identically is $P = I$, $Q = I$. (If $m = n$ this is the statement that the only change of basis leaving every matrix numerically unchanged is the identity basis.)

5 RREF

The matrix on the handout was printed as a stacked list of nine entries. We interpret it as the 3×3 matrix obtained by reading those entries in row-major order

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 3 & 0 & 1 \\ 6 & 5 & 2 \end{pmatrix}.$$

(The algorithm and structural theorems below do not depend on this choice.)

5.1 Q1. The RREF algorithm, and $\mathrm{RREF}(A)$

Prompt

State the RREF algorithm and compute the RREF of $A = \begin{pmatrix} 0 & 1 & 0 \\ 3 & 0 & 1 \\ 6 & 5 & 2 \end{pmatrix}$.

Course-level solution

Algorithm (Gaussian elimination to reduced row echelon form). Start with A and a "current" top-left position $(r, c) = (1, 1)$. Repeat until you run out of rows or columns:

1. Among rows $r, \ldots, m$, find a row whose entry in column c is nonzero. If none exists, increment c and repeat.
2. Swap that row with row r (a pivot now sits at (r, c)).
3. Scale row r so the pivot equals 1.
4. Add multiples of row r to every other row so that all other entries in column c become 0.
5. Increment r and c and continue.

The output is $\mathrm{RREF}(A)$: each pivot is 1, each pivot is the first nonzero entry of its row, and each pivot column is a standard basis vector.

Computation. Column 1 is nonzero. Swap row 1 with row 2, then scale the new row 1 by $1/3$:

$$\begin{pmatrix} 3 & 0 & 1 \\ 0 & 1 & 0 \\ 6 & 5 & 2 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 0 \\ 6 & 5 & 2 \end{pmatrix}.$$

Subtract 6 times row 1 from row 3:

$$\begin{pmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 0 \\ 0 & 5 & 0 \end{pmatrix}.$$

Subtract 5 times row 2 from row 3:

$$\begin{pmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Pivots are already 1 and already clear above. So

$$\mathrm{RREF}(A) = \begin{pmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

5.2 Q2. $\mathrm{RREF}(A) = GA$ for some invertible G

Prompt

Prove that $\mathrm{RREF}(A) = GA$ for some invertible G.

Course-level solution

Each elementary row operation is left-multiplication by an elementary matrix E (swap: a transposition matrix; scale row i by $\lambda \neq 0$: the diagonal matrix with λ in position

i; add λ times row j to row i: $I + \lambda E_{ij}$). Every elementary matrix is invertible (the inverse operation is elementary). The algorithm applies finitely many such operations, so $\mathrm{RREF}(A) = E_k \cdots E_1 A = GA$ with $G = E_k \cdots E_1$ invertible.

5.3 Q3. $\ker(T_A) = \ker(T_{\mathrm{RREF}(A)})$

Prompt

Prove that $\ker(T_A) = \ker(T_{\mathrm{RREF}(A)})$.

Course-level solution

By Q2, $\mathrm{RREF}(A) = GA$ with G invertible. Then $A\vec{x} = \vec{0}$ iff $GA\vec{x} = \vec{0}$ (multiply by G or by G^{-1}). Thus $\ker T_A = \ker T_{GA} = \ker T_{\mathrm{RREF}(A)}$.

5.4 Q4. Example with $\mathrm{im}(T_A) \neq \mathrm{im}(T_{\mathrm{RREF}(A)})$

Prompt

Find an example of A with $\mathrm{im}(T_A) \neq \mathrm{im}(T_{\mathrm{RREF}(A)})$.

Course-level solution

Row operations change column space in general. Take

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \qquad \mathrm{RREF}(A) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Then $\mathrm{im}(T_A) = \mathrm{Span}\{(1,1)^T\}$ while $\mathrm{im}(T_{\mathrm{RREF}(A)}) = \mathrm{Span}\{(1,0)^T\}$, which are unequal.

5.5 Q5. Same image without A already in RREF

Prompt

Is it possible that $\mathrm{im}(T_A) = \mathrm{im}(T_{\mathrm{RREF}(A)})$ but $A \neq \mathrm{RREF}(A)$?

Course-level solution

Yes. Take

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \mathrm{RREF}(A) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

The column spaces are both $\mathrm{Span}\{(1,0)^T\}$, but $A \neq \mathrm{RREF}(A)$.
(The phenomenon is: scaling a pivot column of A does not change its span, but does change the matrix.)

5.6 Q6. Row-equivalent matrices have the same RREF

Prompt

True/false: If $A = GB$ for some invertible G, then $\mathrm{RREF}(A) = \mathrm{RREF}(B)$.

Course-level solution

True. $A = GB$ with G invertible means A and B are row-equivalent: they have the same row space and, more to the point, each is obtained from the other by elementary row operations. The reduced row echelon form of a matrix is *unique*, so there is only one RREF in a row-equivalence class. Hence $\mathrm{RREF}(A) = \mathrm{RREF}(B)$.
(If uniqueness of RREF has not been proved yet: $\mathrm{RREF}(A) = G_A A = G_A GB$ and $\mathrm{RREF}(B) = G_B B$ are both RREF matrices that are row-equivalent, and one checks that a matrix in RREF is determined by its pivot pattern and row space, or invokes uniqueness.)

Later-course solution (using ideas from the rest of the handout)

Uniqueness of RREF is the key lemma. A quick uniqueness argument: if R and S are both RREF and $R = HS$ for invertible H, then R and S have the same nullspace, hence the same pivot columns (the non-free coordinates), and comparing pivot columns — which are standard basis vectors — forces $H = I$ on the pivot rows, hence $R = S$.

5.7 Q7. RREF when T_A is injective

Prompt

If T_A is injective, what does $\mathrm{RREF}(A)$ look like?

Course-level solution

$T_A : \mathbb{F}^n \to \mathbb{F}^m$ injective means $\ker T_A = \{\vec{0}\}$, so there are no free variables: every column is a pivot column. That requires $n \le m$, and

$$\mathrm{RREF}(A) = \begin{pmatrix} I_n \\ 0_{(m-n)\times n} \end{pmatrix}$$

(the block 0 is absent if $m = n$, in which case $\mathrm{RREF}(A) = I_n$).

5.8 Q8. Bases of $\mathrm{im}(T_A)$ and $\ker(T_A)$ from $\mathrm{RREF}(A)$

Prompt

Given $\mathrm{RREF}(A)$, describe how to find a basis for $\mathrm{im}(T_A)$ and $\ker(T_A)$.

Course-level solution

Let $R = \mathrm{RREF}(A)$.

- **Image / column space of A.** The pivot columns of R mark the pivot columns of A.

Those columns of the *original* matrix A form a basis of $\mathrm{im}(T_A)$. (Do not take columns of R: Q4 shows they may span the wrong space.)

- **Kernel / nullspace of A.** Free variables correspond to non-pivot columns. For each free variable x_j, set $x_j = 1$ and set the other free variables to 0; solve the pivot equations (which are immediate from R) for the pivot variables. The resulting n special solutions form a basis of $\ker(T_A)$. Equivalently, if the non-pivot columns of R are $R\vec{e}_{j_k}$, each special solution is $\vec{e}_{j_k}$ minus the linear combination of pivot columns that R uses to express that column.

Illustration with the matrix of Q1. $R = \begin{pmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ has pivots in columns $1, 2$. A basis of $\mathrm{im}(T_A)$ is the first two columns of A: $(0, 3, 6)^T$ and $(1, 0, 5)^T$. One free variable x_3, and $x_1 + (1/3)x_3 = 0$, $x_2 = 0$, so a basis of the kernel is $\big((-1/3),\, 0,\, 1\big)^T$.

6 Fundamental theorems

6.1 Q1. State the fundamental theorems of linear algebra

Prompt

State the fundamental theorems of linear algebra.

Course-level solution

At the level of this midterm (finite-dimensional spaces, no inner products required), the core theorems are the following.

Theorem (Rank–nullity)**.** If $T : V \to W$ is linear and $\dim V < \infty$, then $\dim \mathrm{im}(T) + \dim \ker(T) = \dim V$.

Theorem (Rank)**.** For $A \in \mathbb{F}^{m \times n}$,

$$\mathrm{rank}(A) := \dim \mathrm{im}(T_A) = (\text{number of pivots in } \mathrm{RREF}(A))$$

equals the dimension of the column space and of the row space of A.

Theorem (Characterizations of injectivity / surjectivity)**.** Let $T : V \to W$ with $\dim V = n < \infty$, $\dim W = m < \infty$, and let $\mathcal{B}$ be a basis of V.

- T injective $\iff$ $\ker T = \{\vec{0}\}$ $\iff$ $T(\mathcal{B})$ independent $\iff$ $\dim \mathrm{im}(T) = n$ $\iff$ $n \le m$ and rank $= n$.
- T surjective $\iff$ $\mathrm{im}(T) = W$ $\iff$ $T(\mathcal{B})$ spans W $\iff$ $\dim \mathrm{im}(T) = m$ $\iff$ $m \le n$ and rank $= m$.
- If $n = m$, these are all equivalent to T being invertible.

For T_A, injectivity means every column is a pivot; surjectivity means every row has a pivot.

Theorem (Left and right inverses)**.** T_A is injective iff A has a left inverse ($BA = I_n$ for some B). T_A is surjective iff A has a right inverse ($AC = I_m$ for some C).

(If the course has already named the "four subspaces" theorem over $\mathbb{R}$ with orthogonality, that is a later refinement using a transpose/inner product and is not needed for the proofs below.)

6.2 Q2. T_A injective implies A has a left inverse

Prompt

Prove that T_A injective implies A has a left inverse.

Course-level solution

Let $A \in \mathbb{F}^{m\times n}$ with $T_A : \mathbb{F}^n \to \mathbb{F}^m$ injective. Then $\ker T_A = \{\vec{0}\}$, so the columns of A are independent. In particular $n \le m$.
Let $\mathcal{B} = (Ae_1, \dots, Ae_n)$, a linearly independent list in $\mathbb{F}^m$. Extend it to a basis $\mathcal{C} = (\vec{w}_1, \dots, \vec{w}_m)$ of $\mathbb{F}^m$ with $\vec{w}_j = Ae_j$ for $j \le n$ (possible because we may add vectors from a spanning list, e.g. the standard basis, that are not yet in the span).
Define $S : \mathbb{F}^m \to \mathbb{F}^n$ to be the unique linear map with

$$S(\vec{w}_j) = \begin{cases} e_j & 1 \le j \le n, \\ \vec{0} & n < j \le m. \end{cases}$$

Then $S(T_A(e_j)) = S(\vec{w}_j) = e_j$ for $j = 1, \dots, n$, so $S \circ T_A = \mathrm{id}_{\mathbb{F}^n}$. Letting B be the standard matrix of S, we have $BA = I_n$.

Later-course solution (using ideas from the rest of the handout)

By rank-nullity, injectivity means $\mathrm{rank}(A) = n$. Then $\mathrm{RREF}(A) = \begin{pmatrix} I_n \\ 0 \end{pmatrix} = GA$ for invertible G, so the first n rows of G form a left inverse of A. This is the computational version of the same fact.

6.3 Q3. T_A surjective implies A has a right inverse

Prompt

Prove that T_A surjective implies A has a right inverse.

Course-level solution

Let $T_A : \mathbb{F}^n \to \mathbb{F}^m$ be surjective. For each standard basis vector $e_i \in \mathbb{F}^m$ choose a preimage $\vec{v}_i \in \mathbb{F}^n$ with $A\vec{v}_i = e_i$. Let C be the $n \times m$ matrix with columns $\vec{v}_1, \dots, \vec{v}_m$. Then

$$ACe_i = A\vec{v}_i = e_i \qquad (i = 1, \dots, m),$$

so $AC = I_m$. (The map T_C is a right inverse of T_A.)

Later-course solution (using ideas from the rest of the handout)

Surjectivity means $\operatorname{rank}(A) = m$, so $m \le n$ and $\operatorname{RREF}(A) = \begin{pmatrix} I_m & F \end{pmatrix}$ after row operations: $GA = \begin{pmatrix} I_m & F \end{pmatrix}$. A particular right inverse is $C = \begin{pmatrix} I_m \\ 0 \end{pmatrix} G$ when we pad with free-variable zeros after undoing G, or more invariantly: choose any complementary subspace to $\ker T_A$ (Dimension Q6); T_A restricted there is an isomorphism $\mathbb{F}^n/\text{direction of } \ker \to \mathbb{F}^m$, and the inverse of that isomorphism, extended by 0 on a complement, is a right inverse. The preimage-of-basis argument above is the one to write at this point in the recitation.

7 Subspaces

7.1 Q1. Definition

Prompt

Define "V is a subspace of W".

Course-level solution

Definition. A subset $V \subseteq W$ is a *subspace* of the vector space W if V is itself a vector space under the addition and scalar multiplication of W.

In practice one checks the subspace criteria: $\vec{0} \in V$, and $a\vec{u} + b\vec{v} \in V$ whenever $\vec{u}, \vec{v} \in V$ and $a, b \in \mathbb{F}$ (equivalently: closed under addition and under scalar multiplication).

7.2 Q2. Kernel and image are subspaces

Prompt

Show that $\ker(T)$ and $\operatorname{im}(T)$ are subspaces if T is a linear transformation.

Course-level solution

Let $T : V \to W$ be linear.
Kernel. $T(\vec{0}) = \vec{0}$, so $\vec{0} \in \ker T$. If $\vec{v}, \vec{w} \in \ker T$ and $a, b \in \mathbb{F}$, then

$$T(a\vec{v} + b\vec{w}) = aT(\vec{v}) + bT(\vec{w}) = \vec{0},$$

so $a\vec{v} + b\vec{w} \in \ker T$.
Image. $\vec{0} = T(\vec{0}) \in \operatorname{im} T$. If $\vec{w}_i = T(\vec{v}_i)$ and $a, b \in \mathbb{F}$, then $a\vec{w}_1 + b\vec{w}_2 = T(a\vec{v}_1 + b\vec{v}_2) \in \operatorname{im} T$.

7.3 Q3. Without linearity, those sets need not be subspaces

Prompt

Show that if f is not a linear transformation, $\{\vec{v} \in V : f(\vec{v}) = \vec{0}\}$ and $\{f(\vec{v}) : \vec{v} \in V\}$ need not be subspaces.

Course-level solution

Take $V = W = \mathbb{R}$ over $\mathbb{R}$.
Zero-set need not be a subspace. Let $f(x) = x^2 - 1$. Then $\{x : f(x) = 0\} = \{-1, 1\}$, which does not contain 0 and is not closed under addition.
(Even simpler: $f(x) = x + 1$. Then $\{x : f(x) = 0\} = \{-1\}$, not a subspace.)
Image need not be a subspace. Let $g(x) = x^2$. Then $\{g(x) : x \in \mathbb{R}\} = [0, \infty)$, which is not closed under scalar multiplication by -1.
(Alternatively $g(x) = 1$ constantly: the image is $\{1\}$, which does not contain 0.)

7.4 Q4. Images and preimages of subspaces

Prompt

For all subspaces $U_1 \subseteq V$, $U_2 \subseteq W$ and $T : V \to W$, we have that $T^{-1}(U_2) \subseteq V$ and $T(U_1) \subseteq W$ are subspaces.

Course-level solution

Assume T is linear (as is implicit in this section of the course).
Preimage. $T(\vec{0}) = \vec{0} \in U_2$, so $\vec{0} \in T^{-1}(U_2)$. If $\vec{v}, \vec{w} \in T^{-1}(U_2)$ and $a, b \in \mathbb{F}$, then $T(\vec{v}), T(\vec{w}) \in U_2$, hence $T(a\vec{v} + b\vec{w}) = aT(\vec{v}) + bT(\vec{w}) \in U_2$ because U_2 is a subspace. Thus $a\vec{v} + b\vec{w} \in T^{-1}(U_2)$.
Image. $T(\vec{0}) = \vec{0} \in T(U_1)$. If $T(\vec{v}), T(\vec{w})$ are in $T(U_1)$ with $\vec{v}, \vec{w} \in U_1$, then $a\vec{v} + b\vec{w} \in U_1$, so $aT(\vec{v}) + bT(\vec{w}) = T(a\vec{v} + b\vec{w}) \in T(U_1)$.

Later-course solution (using ideas from the rest of the handout)

These are the statements $\ker T = T^{-1}(\{\vec{0}\})$ and $\operatorname{im} T = T(V)$ as special cases. After rank-nullity one also knows dimensions: $\dim T(U_1) = \dim U_1 - \dim(U_1 \cap \ker T)$ and $\dim T^{-1}(U_2) = \dim(U_2 \cap \operatorname{im} T) + \dim \ker T$.

8 Dimension

8.1 Q1. An independent list is no longer than a basis

Prompt

Prove that the length of a linearly independent list is at most the length of a basis.

Course-level solution

Let $(\vec{v}_1, \ldots, \vec{v}_k)$ be linearly independent in V, and let $(\vec{w}_1, \ldots, \vec{w}_n)$ be a basis of V. We show $k \leq n$ by the replacement (Steinitz) argument.
The list $(\vec{v}_1, \vec{w}_1, \ldots, \vec{w}_n)$ is spanning (it still spans V) and linearly dependent ($\vec{v}_1$ is in the span of the $\vec{w}$'s). In a dependent spanning list, some vector is in the span of the preceding ones; that vector cannot be $\vec{v}_1$ (independence of a single nonzero vector, or $\vec{v}_1 \neq \vec{0}$), so some $\vec{w}_{j_1}$ is in the span of $\{\vec{v}_1\} \cup \{\vec{w}_i : i \neq j_1\}$. Drop it: $(\vec{v}_1, \vec{w}_1, \ldots, \widehat{\vec{w}_{j_1}}, \ldots, \vec{w}_n)$ still spans.
Repeat: adjoin $\vec{v}_2$ on the left. The list is dependent, and sequential independence of $(\vec{v}_1, \vec{v}_2)$ means the vector we drop is one of the remaining $\vec{w}$'s. After k steps we have replaced k of the $\vec{w}$'s by $\vec{v}_1, \ldots, \vec{v}_k$ and still have a spanning list. We cannot have run out of $\vec{w}$'s before using all $\vec{v}$'s: otherwise $(\vec{v}_1, \ldots, \vec{v}_k)$ would be dependent. Hence $k \leq n$.

Later-course solution (using ideas from the rest of the handout)

After linear maps: independence of $(\vec{v}_1, \ldots, \vec{v}_k)$ means the map $\mathbb{F}^k \to V$, $e_i \mapsto \vec{v}_i$ is injective, so $k = \dim \mathbb{F}^k \leq \dim V = n$ once one knows injective maps cannot raise dimension (Dimension Q3 / rank-nullity). That proof is shorter but logically later.

8.2 Q2. Define $\dim(V)$

Prompt

Define $\dim(V)$.

Course-level solution

Definition. If V has a finite basis, the *dimension* $\dim V$ is the length of any basis of V. This is well-defined by Q1: two bases are independent lists each spanning, so each is no longer than the other.
If V has no finite spanning list, one writes $\dim V = \infty$. $\dim\{\vec{0}\} = 0$, the length of the empty basis.

8.3 Q3. Injective maps preserve dimension of the image

Prompt

Show that if $T : V \to W$ is an injective linear transformation, $\dim(\mathrm{im}(T)) = \dim(V)$.

Course-level solution

Assume $\dim V = n < \infty$ (otherwise both sides are infinite if T is injective and V is infinite-dimensional, with the usual caveats). Let $(\vec{v}_1, \ldots, \vec{v}_n)$ be a basis of V. Linear Transformations Q6 says $\big(T(\vec{v}_1), \ldots, T(\vec{v}_n)\big)$ is independent; Linear Transformations Q5 says it spans $\mathrm{im}(T)$. So it is a basis of $\mathrm{im}(T)$ of length n, and $\dim \mathrm{im}(T) = n = \dim V$.
(Equivalently: $T : V \to \mathrm{im}(T)$ is an isomorphism, and isomorphisms preserve the length of bases.)

Later-course solution (using ideas from the rest of the handout)

Rank-nullity: $\dim \operatorname{im}(T) = \dim V - \dim \ker T = \dim V$ because $\ker T = \{\vec{0}\}$.

8.4 Q4. Extension lemma

Prompt

Extension Lemma: If $\dim(V) < \infty$, any linearly independent list can be extended to a basis.

Course-level solution

Let $(\vec{v}_1, \ldots, \vec{v}_k)$ be independent and let $(\vec{w}_1, \ldots, \vec{w}_n)$ be a basis, $n = \dim V$. Consider the concatenated list $(\vec{v}_1, \ldots, \vec{v}_k, \vec{w}_1, \ldots, \vec{w}_n)$. It spans V. Apply the extraction procedure of Span Q5: keep a vector iff it is not in the span of those already kept. All of $\vec{v}_1, \ldots, \vec{v}_k$ are kept (they start independent). The surviving list is independent, still spans V, and is therefore a basis extending the original list.

Later-course solution (using ideas from the rest of the handout)

If you already know that independent lists have length $\leq n$ and that a spanning list of length n is a basis, you may add one vector at a time from outside the current span; the process stops by length at latest at n, and must stop with a spanning list.

8.5 Q5. Dimension of a subspace

Prompt

If $V \subseteq W$ is a subspace, $\dim(V) \leq \dim(W)$.

Course-level solution

If $\dim W = \infty$ there is nothing to prove. If $\dim W = n < \infty$, a basis of V is a linearly independent list in W, hence has length at most n by Q1.
(If $\dim V = \dim W < \infty$ then any basis of V is an independent list of length $\dim W$ in W, hence a basis of W, so $V = W$.)

8.6 Q6. Complementary subspace

Prompt

If $\dim(W) < \infty$ and $U \subseteq W$ is a subspace, then there exists a subspace $V \subseteq W$ such that $U \cap V = \{\vec{0}\}$ and $\dim(U) + \dim(V) = \dim(W)$.

Course-level solution

Let $(\vec{u}_1, \ldots, \vec{u}_k)$ be a basis of U. Extend it to a basis $(\vec{u}_1, \ldots, \vec{u}_k, \vec{v}_1, \ldots, \vec{v}_{n-k})$ of W by the extension lemma, where $n = \dim W$. Set $V = \operatorname{Span}(\vec{v}_1, \ldots, \vec{v}_{n-k})$.

Then $\dim V = n-k$, so $\dim U + \dim V = n = \dim W$. If $\vec{x} \in U \cap V$, then $\vec{x} = \sum a_i \vec{u}_i = \sum b_j \vec{v}_j$, hence $\sum a_i \vec{u}_i - \sum b_j \vec{v}_j = \vec{0}$. Independence of the full basis forces all coefficients to vanish, so $\vec{x} = \vec{0}$.

Later-course solution (using ideas from the rest of the handout)

V is a complement of U in W: equivalently $W = U \oplus V$. Complements are not unique unless $U = W$ or $U = \{\vec{0}\}$. After you have an inner product one can choose the orthogonal complement, which is unique; that is later than this handout.

8.7 Q7. $\dim(U + V) = \dim U + \dim V - \dim(U \cap V)$

Prompt

Show that $\dim(U + V) = \dim(U) + \dim(V) - \dim(U \cap V)$.

Course-level solution

Assume $U, V \subseteq W$ are finite-dimensional (e.g. if $\dim W < \infty$). Let $(\vec{z}_1, \ldots, \vec{z}_r)$ be a basis of $U \cap V$. Extend it to a basis $(\vec{z}_1, \ldots, \vec{z}_r, \vec{u}_1, \ldots, \vec{u}_s)$ of U and to a basis $(\vec{z}_1, \ldots, \vec{z}_r, \vec{v}_1, \ldots, \vec{v}_t)$ of V. We claim

$$\mathcal{B} = (\vec{z}_1, \ldots, \vec{z}_r, \vec{u}_1, \ldots, \vec{u}_s, \vec{v}_1, \ldots, \vec{v}_t)$$

is a basis of $U + V$.

Spanning. A typical vector of $U + V$ is $\vec{u} + \vec{v}$ with $\vec{u} \in U$, $\vec{v} \in V$, hence is a combination of the $\vec{z}$'s, $\vec{u}$'s, and $\vec{v}$'s.

Independence. If $\sum a_i \vec{z}_i + \sum b_j \vec{u}_j + \sum c_k \vec{v}_k = \vec{0}$, then $\sum c_k \vec{v}_k = -\sum a_i \vec{z}_i - \sum b_j \vec{u}_j \in U$, so $\sum c_k \vec{v}_k \in U \cap V = \mathrm{Span}(\vec{z}_i)$. But $(\vec{z}_1, \ldots, \vec{z}_r, \vec{v}_1, \ldots, \vec{v}_t)$ is independent, so every $c_k = 0$ and then every $a_i, b_j = 0$ by independence of the U-basis.

Thus $\dim(U + V) = r + s + t$, while $\dim U = r + s$, $\dim V = r + t$, $\dim(U \cap V) = r$, which is the formula.

Later-course solution (using ideas from the rest of the handout)

Map $U \times V \to U + V$ by $(\vec{u}, \vec{v}) \mapsto \vec{u} + \vec{v}$. It is surjective with kernel $\{(x, -x) : x \in U \cap V\} \cong U \cap V$. Rank-nullity on $U \times V$ (whose dimension is $\dim U + \dim V$) yields the same identity. This is the slick end-of-handout proof.

9 Rank–nullity

9.1 Q1. $\dim \mathrm{im}(T_A) = \mathrm{rank}(A)$

Prompt

Prove that $\dim(\mathrm{im}(T_A)) = \mathrm{rank}(A)$.

Course-level solution

By definition, $\mathrm{im}(T_A)$ is the column space of A, and $\mathrm{rank}(A)$ is defined to be the dimension of the column space (equivalently, the number of pivot columns of A, which form a basis of that column space by RREF Q8). Hence the equality is tautological once those identifications are made.
If rank was defined as the number of nonzero rows of $\mathrm{RREF}(A)$: that number equals the number of pivot columns, which is the length of a basis of $\mathrm{im}(T_A)$.

9.2 Q2. Rank–nullity

Prompt

Rank-nullity. Prove that if $T : V \to W$ and $\dim(V) < \infty$, $\dim(\mathrm{im}(T)) + \dim(\ker(T)) = \dim(V)$.

Course-level solution

Let $n = \dim V$ and let $(\vec{v}_1, \dots, \vec{v}_k)$ be a basis of $\ker T$ (possibly $k = 0$). Extend to a basis $(\vec{v}_1, \dots, \vec{v}_k, \vec{u}_1, \dots, \vec{u}_{n-k})$ of V (extension lemma).
We claim $\big(T(\vec{u}_1), \dots, T(\vec{u}_{n-k})\big)$ is a basis of $\mathrm{im}\, T$.
Spanning. For $\vec{v} = \sum a_i \vec{v}_i + \sum b_j \vec{u}_j$,

$$T(\vec{v}) = \sum b_j T(\vec{u}_j)$$

because $T(\vec{v}_i) = \vec{0}$. So the $T(\vec{u}_j)$ span $\mathrm{im}\, T$.
Independence. If $\sum b_j T(\vec{u}_j) = \vec{0}$, then $T\big(\sum b_j \vec{u}_j\big) = \vec{0}$, so $\sum b_j \vec{u}_j \in \ker T = \mathrm{Span}(\vec{v}_1, \dots, \vec{v}_k)$. Then $\sum b_j \vec{u}_j - \sum c_i \vec{v}_i = \vec{0}$ for some c_i, and independence of the full basis of V forces every $b_j = 0$.
Therefore $\dim \mathrm{im}\, T = n - k = \dim V - \dim \ker T$.

Later-course solution (using ideas from the rest of the handout)

For T_A, this is "number of columns = number of pivot columns + number of free columns", which is visible from RREF. The abstract proof above is the same counting after choosing coordinates.

9.3 Q3. $\dim T^{-1}(U) \le \dim U + \dim \ker T$

Prompt

Let $T : V \to W$ and $U \subseteq W$ a subspace. Show that $\dim(T^{-1}(U)) \le \dim(U) + \dim(\ker(T))$. Find examples of when equality does and doesn't hold.

Course-level solution

Assume $\dim V < \infty$ (so all dimensions below are finite). Restrict the codomain: the restriction

$$T\big|_{T^{-1}(U)} : T^{-1}(U) \longrightarrow U \cap \mathrm{im}(T)$$

is linear, surjective (by definition of image), and has kernel $\ker T \cap T^{-1}(U) = \ker T$. Rank-nullity therefore gives

$$\dim T^{-1}(U) = \dim\big(U \cap \mathrm{im}(T)\big) + \dim \ker T \leq \dim U + \dim \ker T,$$

using $U \cap \mathrm{im}(T) \subseteq U$.
Equality holds if and only if $U \cap \mathrm{im}(T) = U$, i.e. iff $U \subseteq \mathrm{im}(T)$.
Equality example. Let $T = \mathrm{id}_V : V \to V$ and let $U \subseteq V$ be any subspace. Then $\ker T = \{\vec{0}\}$ and $T^{-1}(U) = U$, so $\dim T^{-1}(U) = \dim U = \dim U + \dim \ker T$.
Strict inequality example. Let $T = 0 : V \to W$ with $\dim V \geq 1$ and let $U \subseteq W$ satisfy $\dim U \geq 1$ (e.g. $U = W$ if $\dim W \geq 1$). Then $T^{-1}(U) = V$, and

$$\dim V \leq \dim U + \dim V$$

is equality only if $\dim U = 0$. For $\dim U > 0$ we have a strict inequality: $\dim T^{-1}(U) = \dim V < \dim U + \dim V$.
A mixed example: inclusion $T : \mathbb{F}^2 \to \mathbb{F}^3$, $T(x, y) = (x, y, 0)$, and $U = \mathrm{Span}\{e_3\}$. Then $U \cap \mathrm{im}\, T = \{\vec{0}\}$, $T^{-1}(U) = \ker T = \{\vec{0}\}$, and $0 < 1 + 0$.

Later-course solution (using ideas from the rest of the handout)

The identity $\dim T^{-1}(U) = \dim(U \cap \mathrm{im}\, T) + \dim \ker T$ is the precise sharpening; the inequality in the prompt is what you get by crudely estimating $\dim(U \cap \mathrm{im}\, T) \leq \dim U$.

9.4 Q4. $\dim T(U) \leq \dim U$, and equality

Prompt

Let $T : V \to W$ and $U \subseteq V$ a subspace. Show that $\dim(T(U)) \leq \dim(U)$. When does equality hold?

Course-level solution

The restriction $T|_U : U \to T(U)$ is linear and surjective, with kernel $U \cap \ker T$. Rank-nullity yields

$$\dim T(U) = \dim U - \dim(U \cap \ker T) \leq \dim U.$$

Equality holds iff $\dim(U \cap \ker T) = 0$ iff $U \cap \ker T = \{\vec{0}\}$ iff $T|_U$ is injective.
Special cases: equality for every U iff $\ker T = \{\vec{0}\}$ iff T is injective. Equality for $U = V$ is the statement $\dim \mathrm{im}\, T = \dim V$, again injectivity of T.